Industrial IoT Monitoring using ESP32 + Siemens S7-1200 PLC

1. Introduction

This project focuses on real-time industrial monitoring by integrating Siemens S7-1200 PLC with ESP32 and Firebase Cloud. The main objective was to collect machine and process data from industrial systems and make it available remotely for monitoring, analysis, and decision-making.

In many industries, machine monitoring is still dependent on local supervision and manual observation. This creates delays in identifying issues and reduces operational efficiency. By using Industrial IoT, this project helps improve visibility, remote access, and real-time monitoring of industrial parameters.

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2. Problem Statement

Traditional industrial monitoring systems often require operators to be physically present near the machine or control panel. This leads to:

- Delayed response to machine faults
- Limited real-time visibility
- Increased manual monitoring effort
- Difficulty in accessing production data remotely
- Reduced operational efficiency

There was a need for a system that could provide real-time cloud-based monitoring using existing industrial PLC infrastructure.

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3. Objective

The main objective of this project was to build a cloud-connected industrial monitoring system using Siemens S7-1200 PLC and ESP32.

The system was designed to:

- Collect industrial machine/process data from PLC
- Transfer data using Ethernet TCP/IP communication
- Send the data to Firebase Cloud using ESP32
- Enable remote monitoring of industrial parameters
- Improve industrial visibility and operational control

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4. Technologies Used

Hardware

- Siemens S7-1200 PLC
- ESP32 Development Board
- Ethernet Communication Interface
- Industrial Sensors (process dependent)
- Power Supply Unit

Software

- Embedded C
- Firebase Cloud Platform
- PLC Programming Environment
- TCP/IP Communication Protocol
- Industrial IoT Monitoring System

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5. Working Principle

The industrial machine/process generates operational data such as status, values, and control parameters.

Step-by-Step Flow:

1. Siemens S7-1200 PLC collects machine/process data.
2. The PLC communicates with ESP32 using Ethernet TCP/IP protocol.
3. ESP32 reads the required parameters from the PLC.
4. The ESP32 processes the data and prepares it for cloud transmission.
5. The processed data is sent to Firebase Cloud.
6. Firebase stores the real-time data for remote monitoring.
7. Users can access and monitor the industrial system remotely through cloud access.

Data Flow

Industrial Machine → PLC → Ethernet TCP/IP → ESP32 → Firebase Cloud → Remote Monitoring

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6. My Role

I worked mainly on the ESP32 side of the project and supported the cloud integration process.

My responsibilities included:

- Understanding PLC-to-ESP32 communication flow
- Handling data acquisition from Siemens S7-1200 PLC
- Writing ESP32 logic for data reading and transmission
- Sending collected industrial data to Firebase Cloud
- Supporting stable cloud communication
- Improving reliability for real-time monitoring applications

This helped ensure smooth industrial data transfer and remote monitoring functionality.

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7. Challenges Faced and Troubleshooting

During project development, several practical issues were encountered.

Problem 1: Communication Failure Between PLC and ESP32

Initially, stable communication between Siemens PLC and ESP32 was difficult due to Ethernet communication handling and parameter mapping issues.

Solution

- Verified PLC communication settings
- Checked IP configuration and network setup
- Validated TCP/IP communication flow
- Corrected parameter addressing and data reading logic

This improved communication stability.

Problem 2: Firebase Data Upload Delay

Sometimes data transmission to Firebase was delayed or inconsistent due to network instability.

Solution

- Optimized ESP32 transmission logic
- Improved retry handling for failed uploads
- Reduced unnecessary repeated requests
- Improved network stability checks

This helped maintain smoother cloud updates.

Problem 3: Real-Time Data Synchronization

Mismatch between PLC data updates and cloud update timing created inconsistency.

Solution

- Improved timing control in ESP32 logic
- Added proper data refresh intervals
- Improved synchronization handling

This increased monitoring reliability.

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8. Firebase Result

The final system successfully transmitted industrial PLC data to Firebase Cloud.

Data Successfully Monitored:

- Machine running status
- Process values
- Sensor parameters
- Real-time operational status
- Industrial monitoring variables

Result Achieved

- Real-time remote visibility was established
- Cloud-based monitoring was successful
- Manual supervision requirements were reduced
- Faster monitoring and accessibility improved

The project demonstrated successful Industrial IoT implementation using PLC + ESP32 + Cloud integration.

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9. Benefits

This project provided several industrial advantages:

- Real-time industrial monitoring
- Remote access to machine data
- Reduced manual supervision
- Improved operational visibility
- Better support for future predictive maintenance
- Cloud-based data storage and analysis

- Scalable Industrial IoT architecture

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10. Future Improvements

Future upgrades can make the system even stronger:

- Real-time dashboard visualization
- Mobile app-based monitoring
- Alarm and fault notification system
- Predictive maintenance alerts
- Performance trend analysis
- Advanced industrial analytics
- AI-based fault prediction system

This can convert the project into a complete Industry 4.0 monitoring solution.